



CSC501: Algorithms and Data Models

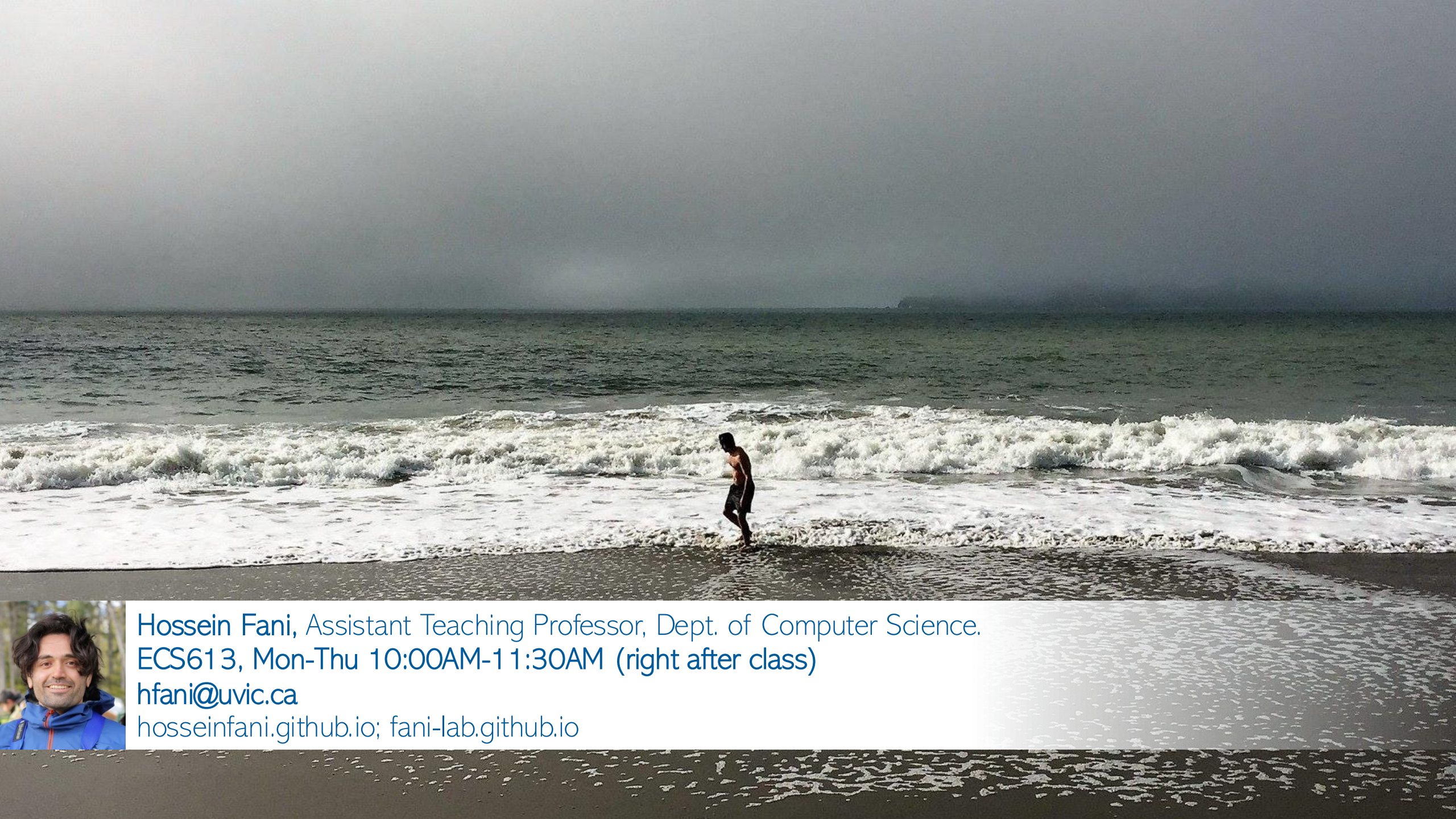
Hossein Fani, Fall 2026



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Graduate Students (30)

Meng in Applied Data Science (21), MSc and PhD in Computer Science (5+3), Master in Management (1)



Hossein Fani, Assistant Teaching Professor, Dept. of Computer Science.
ECS613, Mon-Thu 10:00AM-11:30AM (right after class)

hfani@uvic.ca

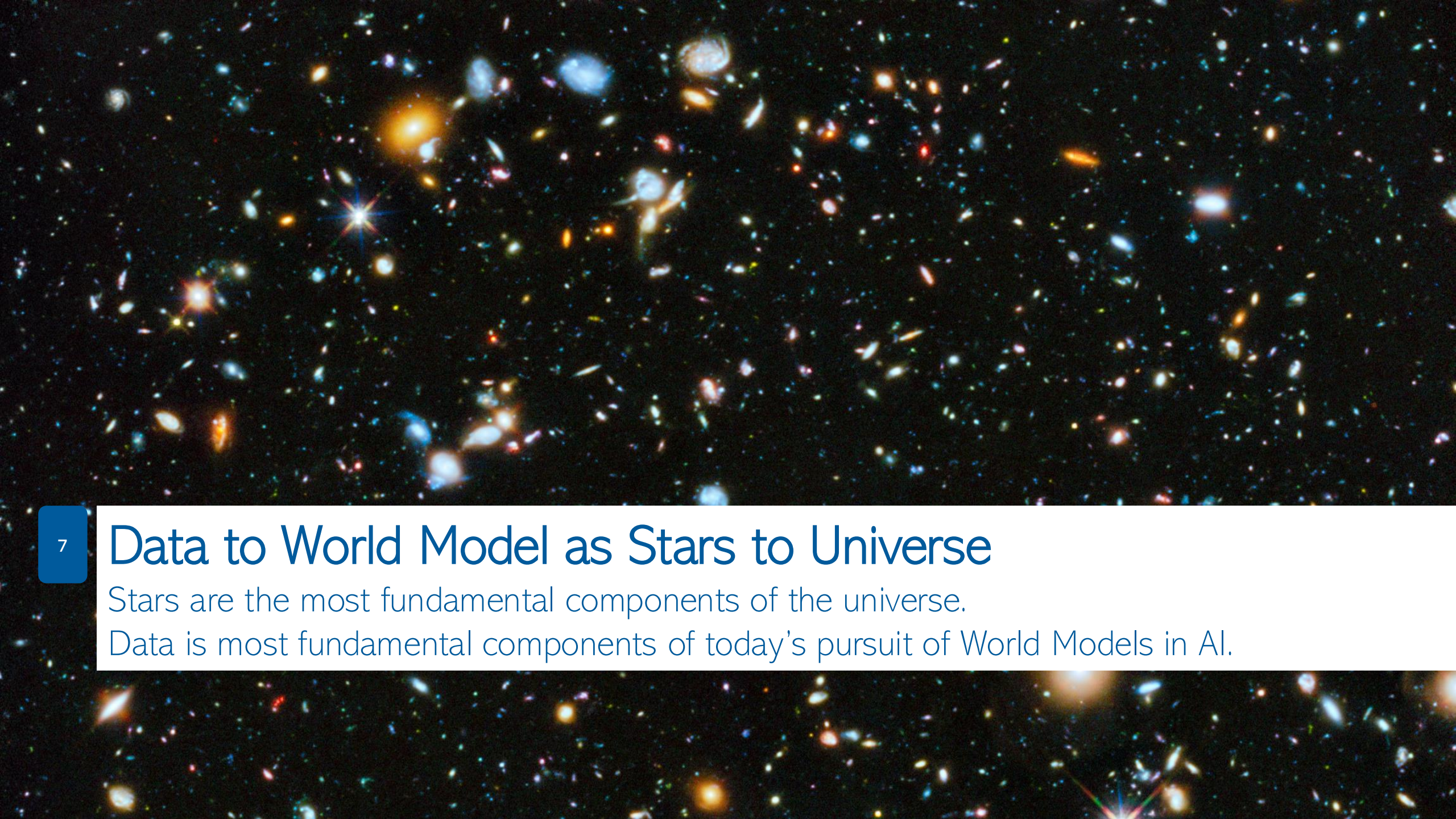
hosseinfani.github.io; fani-lab.github.io



The Constellation Fornax /'fɔːrnæks/

By Hubble Ultra Deep Field, observed 2003, yielding clues to the birth rate of stars over the 13-billion-year universe.
<https://hubblesite.org/contents/media/videos/2014/27/766-Video.html>





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Data to World Model as Stars to Universe

Stars are the most fundamental components of the universe.

Data is most fundamental components of today's pursuit of World Models in AI.

Scoping (Localizing) Data Models

What are we talking about when we talk about Data Models (Modeling Data)?
Stay focused as we zoom in (drill down) through our odyssey ...

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Outline: Week 01 A: Monday, B: Thursday

I) An Odyssey

II) DIKW Pyramid

III) Data Model Abstractions

III) WWW to GGG

Outline: Week 02 A: Monday, Course Logistics & Topics

Outline: W01 B

I) An Odyssey

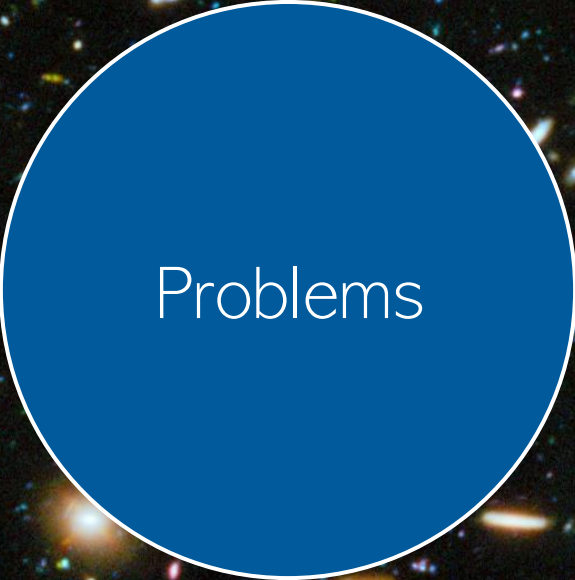
II) DIKW Pyramid

III) Data Model Abstractions

III) WWW to GGG

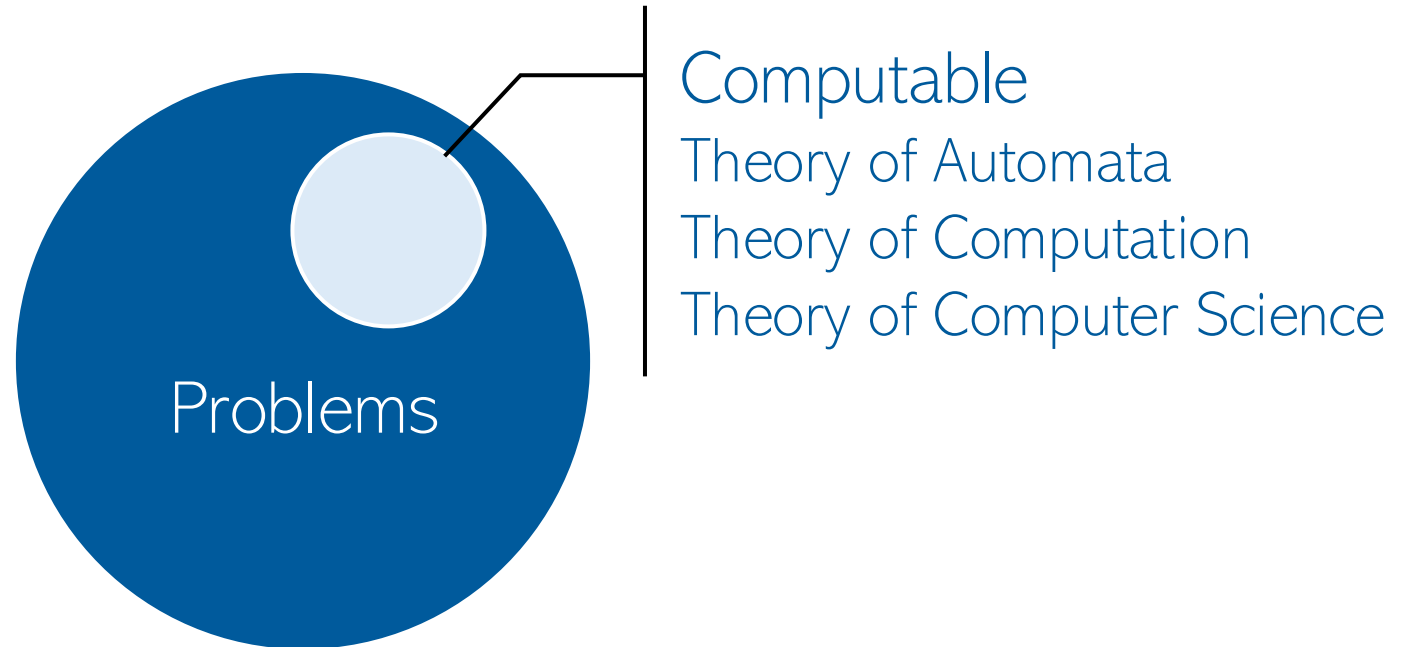
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A Data Odyssey



Problems

A Data Odyssey



Incompleteness: Not all problems have solutions!

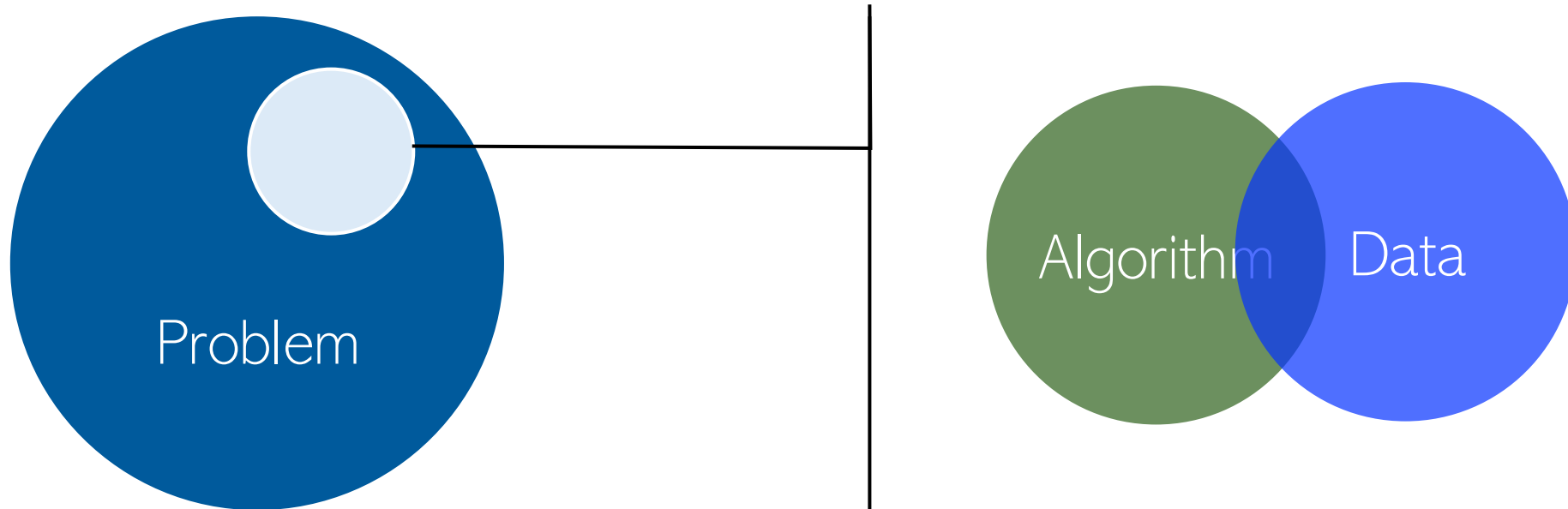
Undecidability: Even if there is a solution, we may not obtain it!

Complexity (Time): Even if we can obtain it, it may take forever (NP)!

Complexity (Space): Even if we can obtain it, it may need huge space!

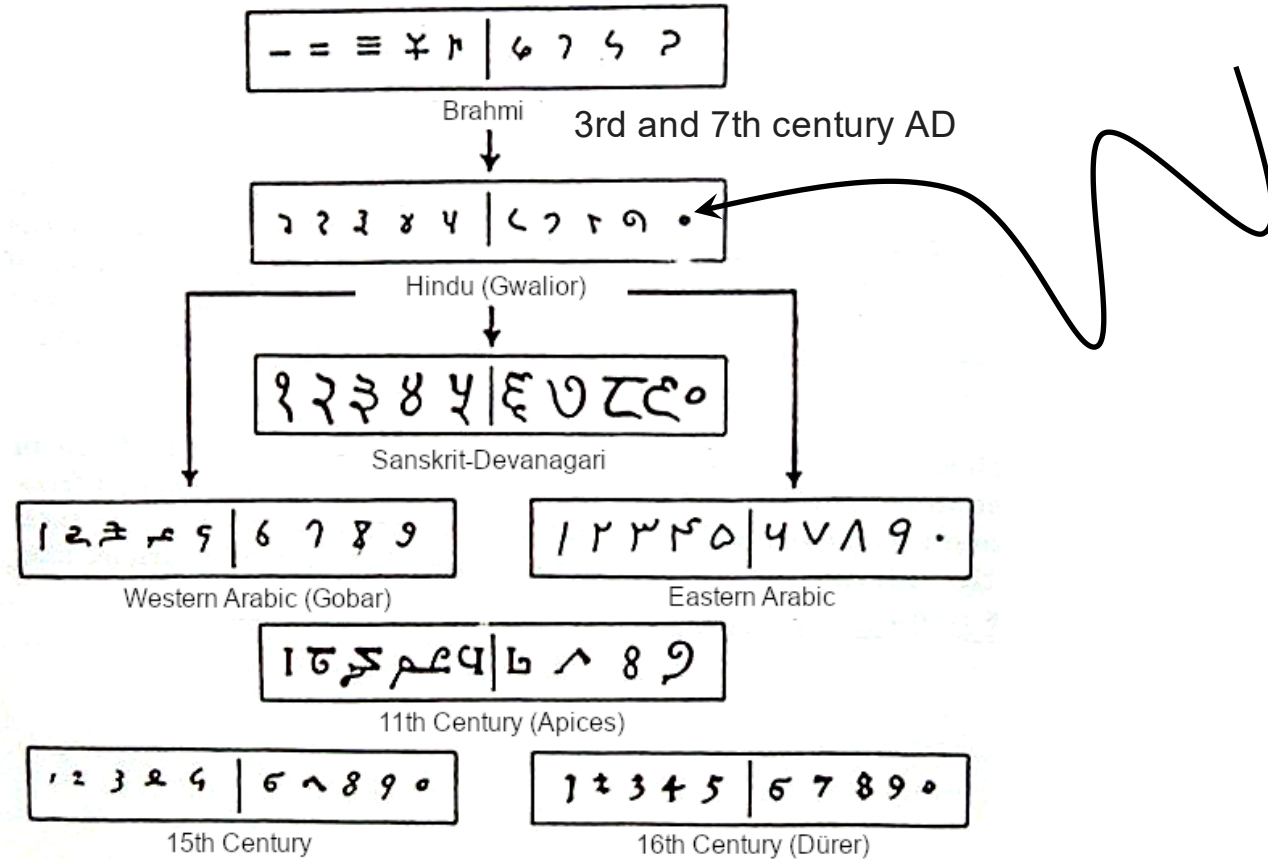
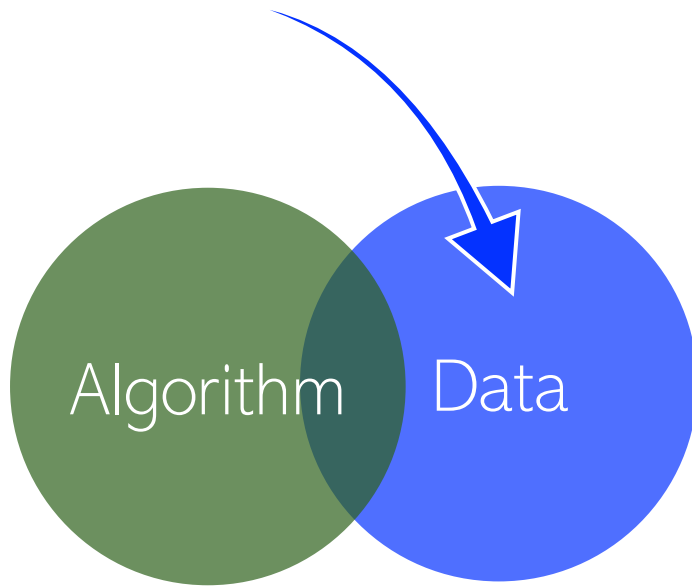
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A Data Odyssey

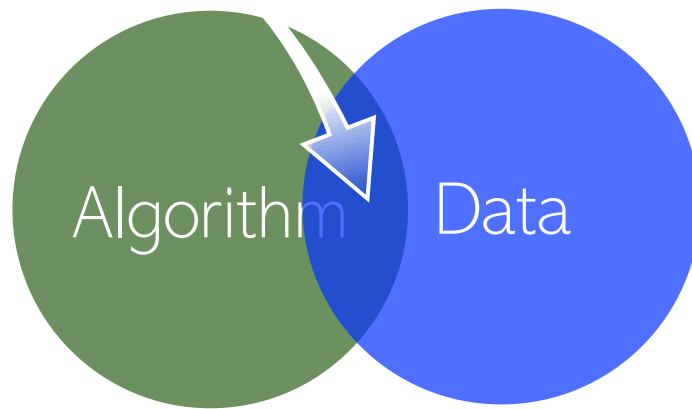


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A Data Odyssey



A Data Odyssey



Data gets bigger!
Algorithms get slower!

Data Engineers $\leftrightarrow$ Algorithm Designers

Better Data Modeling $\leftrightarrow$ Faster Algorithm

- Sorting in Arrays vs. Trees
- Searching in Arrays vs. Trees
- Linear Algebra on Sparse Matrix in Arrays vs. Linked Lists
- ...

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A Data Odyssey

Maya ~750 BC.

A vigesimal (/vɪˈdʒɛsɪməl/) or base-20 numeral system.

https://en.wikipedia.org/wiki/Maya_numerals

0	1	2	3	4
	•	••	•••	••••
5	6	7	8	9
—	•	••	•••	••••
10	11	12	13	14
—	•	••	•••	••••
15	16	17	18	19
—	•	••	•••	••••

400s	
20s	•
1s	•••
	33

	•
	•
	•••
	429

••
•
—
?



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A Data Odyssey

Maya ~750 BC.

A vigesimal (/vɪˈdʒɛsɪməl/) or base-20 numeral system.

https://en.wikipedia.org/wiki/Maya_numerals

$$\text{—} + \text{●●●} = \text{●●●}$$

$$\text{●●●} - \text{—} = \text{●●●}$$



A Data Odyssey

Christina Reynolds-Badder (Digital Design, Fall 2021)

THE VERTICES NUMBER SYSTEM

0, 1, 2, 3, 4, 5

0 → 0	12 → 20
1 → 1	13 → 21
2 → 2	14 → 22
3 → 3	15 → 23
4 → 4	16 → 24
5 → 5	17 → 25
6 → 10	18 → 30
7 → 11	19 → 31
8 → 12	20 → 32
9 → 13	21 → 33
10 → 14	22 → 34
11 → 15	23 → 35

VERTICES → DECIMAL

n = number of positions

V = number of vertices at position

$$\text{decimal} = V \times 6^n + \dots + V \times 6^2 + V \times 6^1 + V \times 6^0$$

ex 1: convert 35 to decimal.

$$3 \times 6^1 + 5 \times 6^0$$

$$= 18 + 5$$

$$= 23 \quad \checkmark$$

ex 2: convert 5542 to decimal.

$$5 \times 6^3 + 5 \times 6^2 + 4 \times 6^1 + 2 \times 6^0$$

$$= 1080 + 180 + 24 + 2$$

$$= 1286 \quad \checkmark$$

DECIMAL → VERTICES

keep dividing by 6, until you can't anymore. The remainder will be the number of vertices, stop when the solution < 1 .

ex convert 23 to vertices.

$$23 / 6 = 3 \text{ with remainder } 5$$

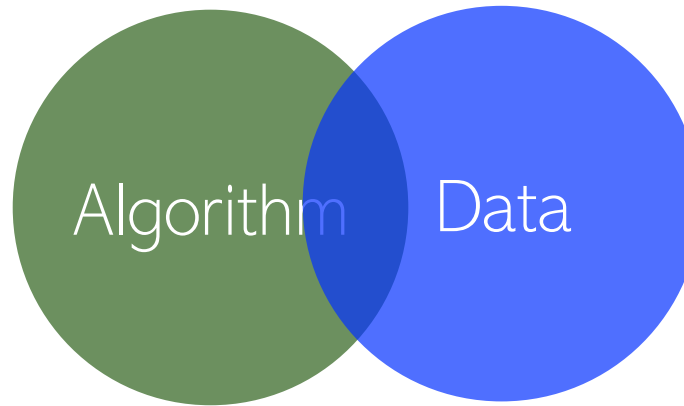
$$3 / 6 = 3 \text{ with remainder } 3$$

3, 5

$$= 35 \quad \checkmark$$

A Data Odyssey

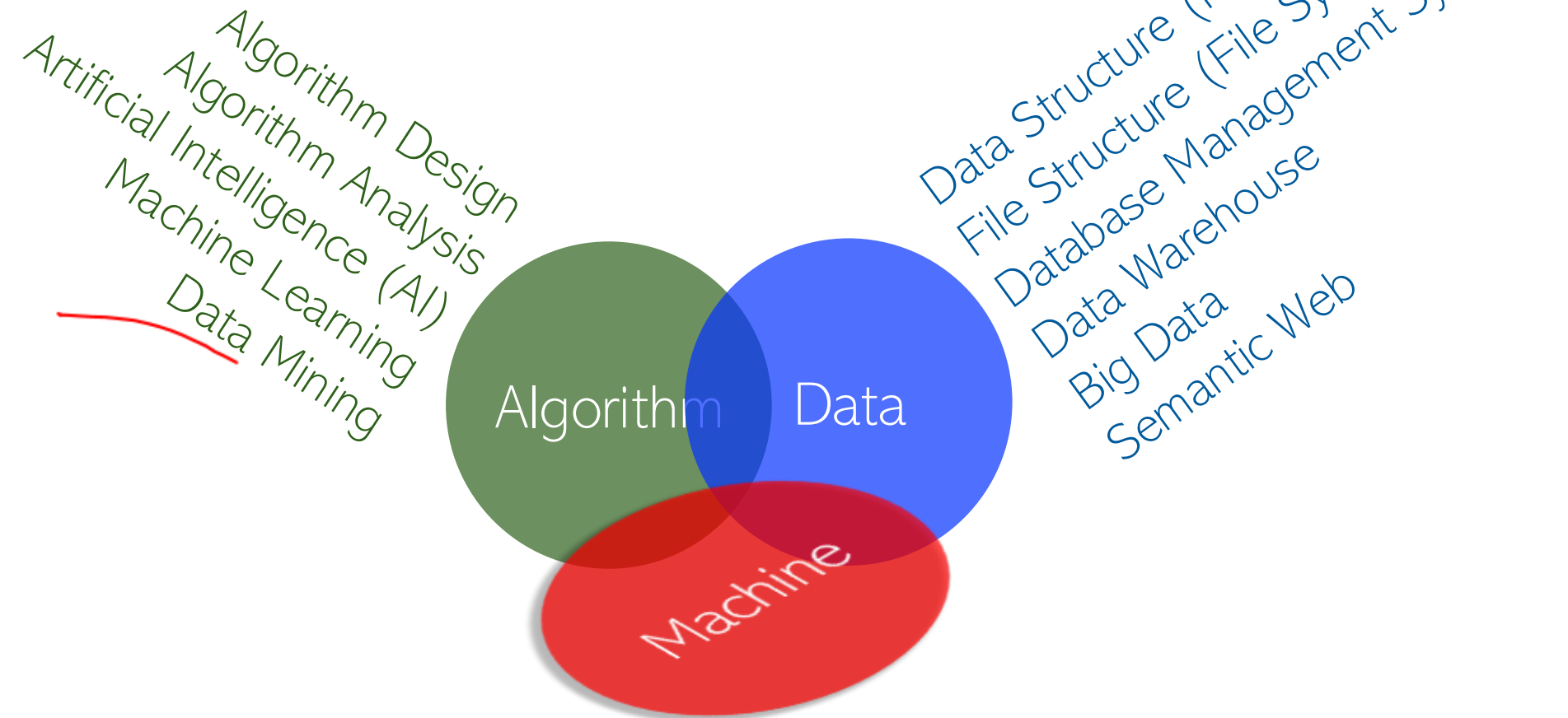
Algorithm Design
Algorithm Analysis
Artificial Intelligence (AI)
Machine Learning
Data Mining



Data Structure (Memory)
File Structure (File Systems)
Database Management Systems
Data Warehouse
Big Data
Semantic Web

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A Data Odyssey



Digital Design (Logic Circuits, qbits)
Computer Architecture (RISC, CISC, CPU, GPU, TPU, QC)
Programming Languages (Assembly, C, Java, Python, ...)
Operating Systems (Unix, Linux, ...)

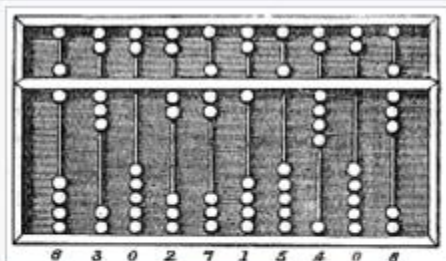
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A Data Odyssey

<https://en.wikipedia.org/wiki/Computer#History>



The Ishango bone, a bone tool dating back to prehistoric Africa.



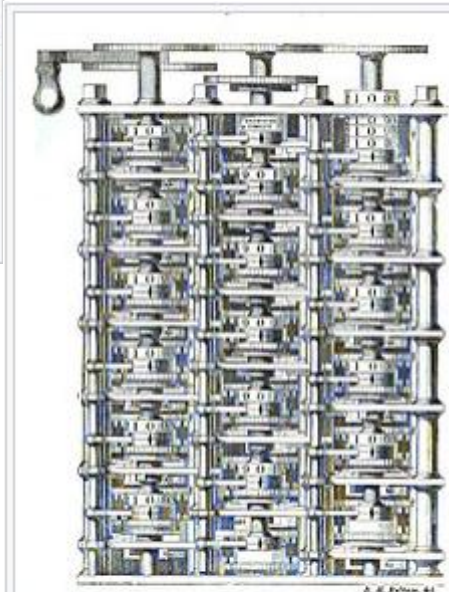
The Chinese **suanpan** (算盘). The number represented on this abacus is 6,302,715,408.

2700 – 2300 BC



The Antikythera mechanism, dating back to ancient Greece circa 150–100 BC, is an early analog computing device.

100 BC: to calculate astronomical positions



A portion of Babbage's Difference engine.

1833: Father of Computer



Replica of Zuse's Z3, the first fully automatic, digital (electromechanical) computer.

1941

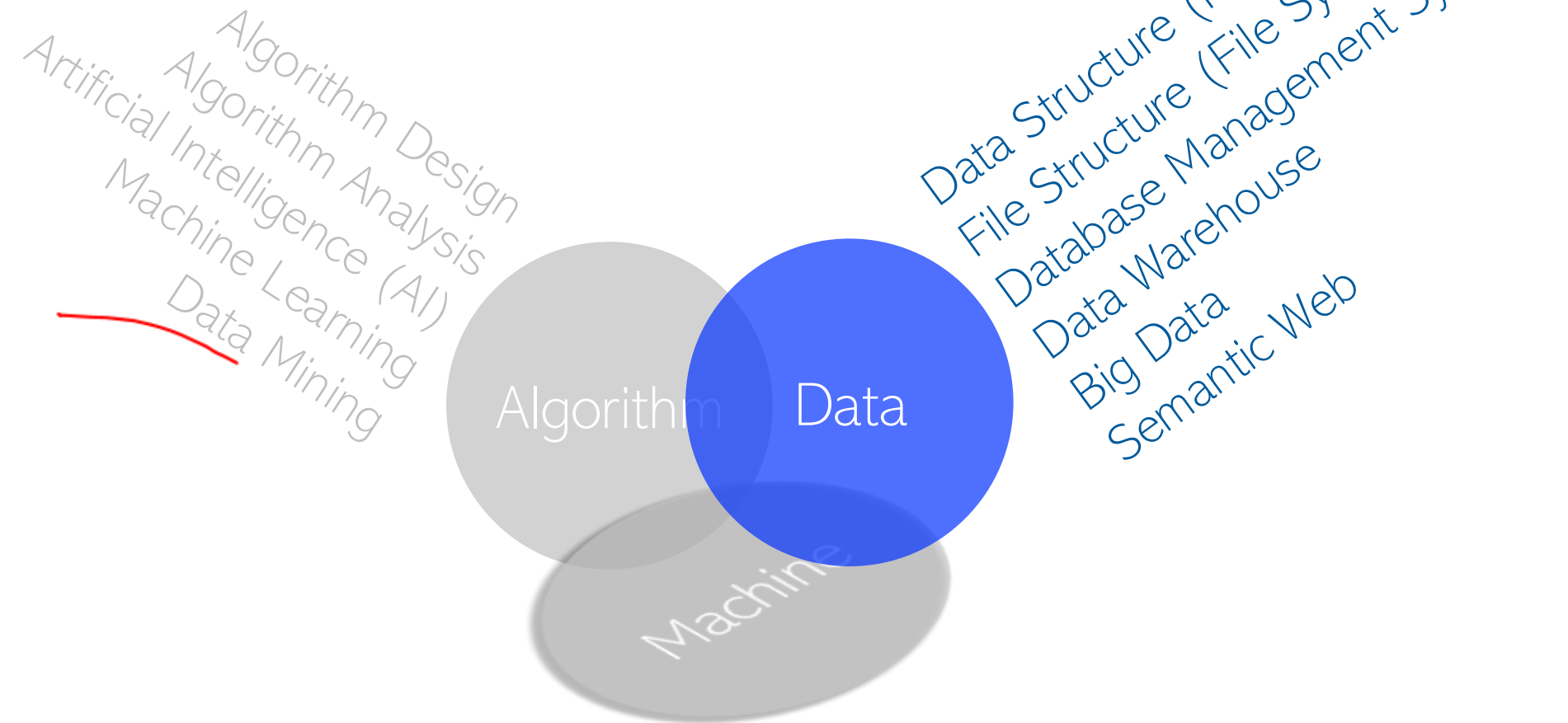
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1943: Colossus

The first electronic digital programmable computing device.
Used to break German ciphers during WWII.

What was the data and data model?

A Data Mining Odyssey



Digital Design (Logic Circuits, qbits)
Computer Architecture (RISC, CISC, CPU, GPU, TPU, QC)
Programming Languages (Assembly, C, Java, Python, ...)
Operating Systems (Unix, Linux, ...)

Outline: W01 B

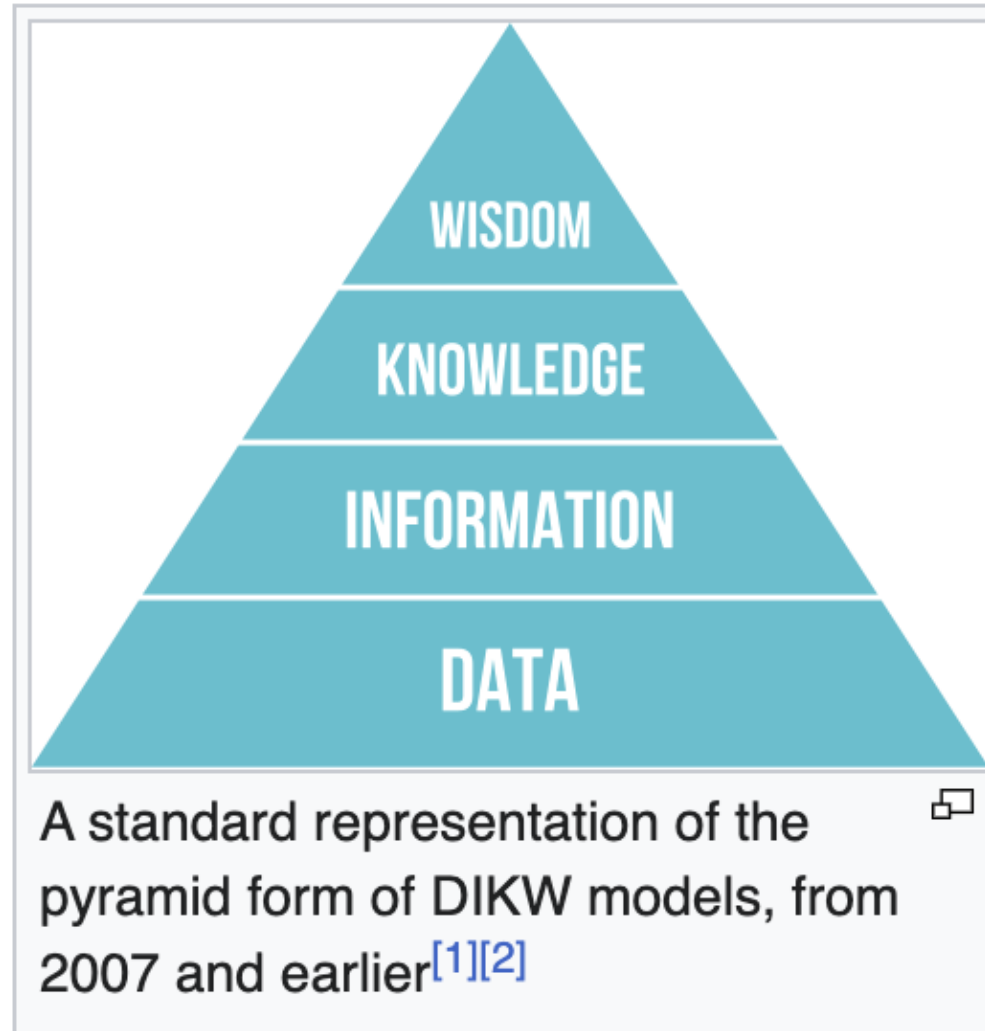
I) An Odyssey

II) DIKW Pyramid

III) Data Model Abstractions

III) WWW to GGG

DIKW Pyramid

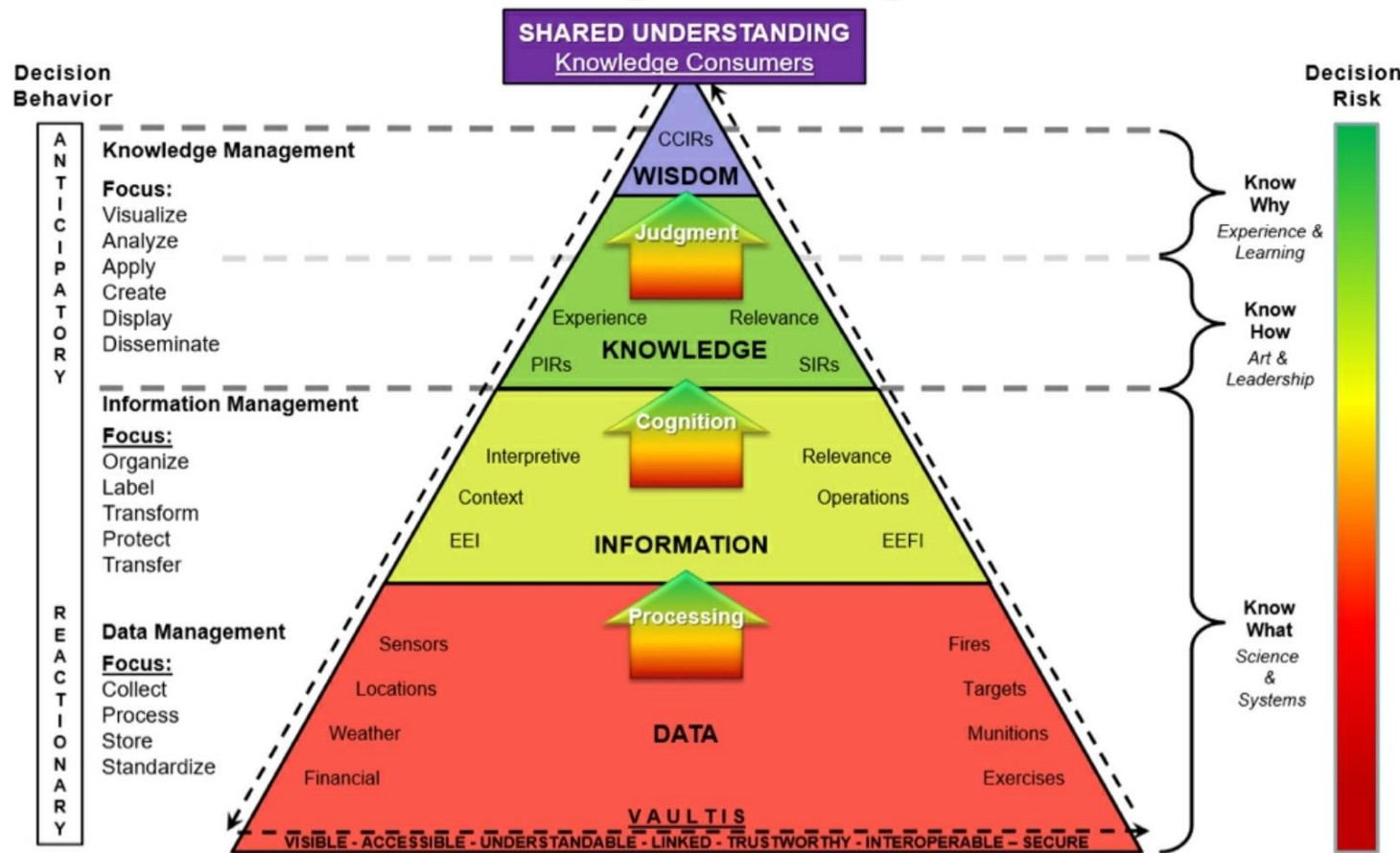


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DIKW Pyramid

The DIKW Pyramid in the U.S. Department of Defense

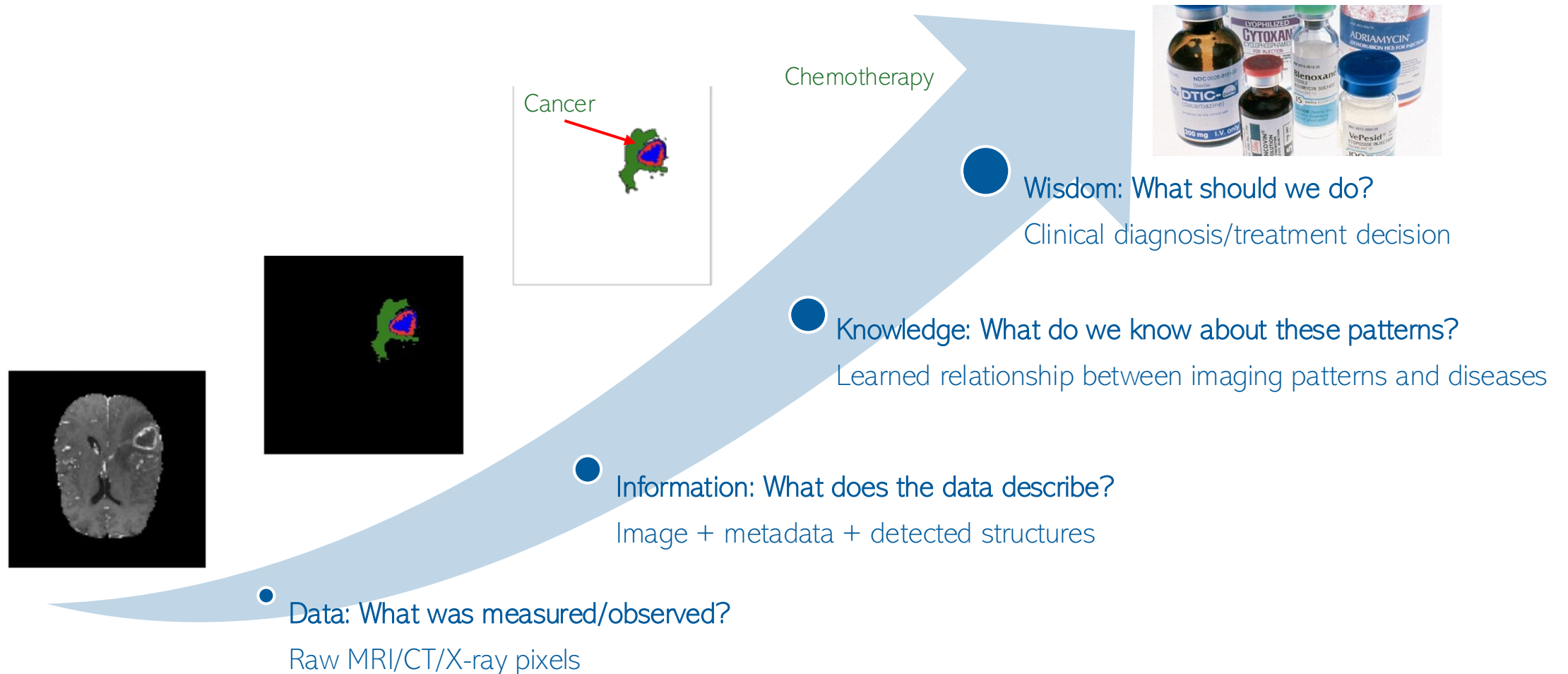
The KM Cognitive Pyramid



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DIKW Pyramid

Example in Biomedical Imaging



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DIKW Pyramid

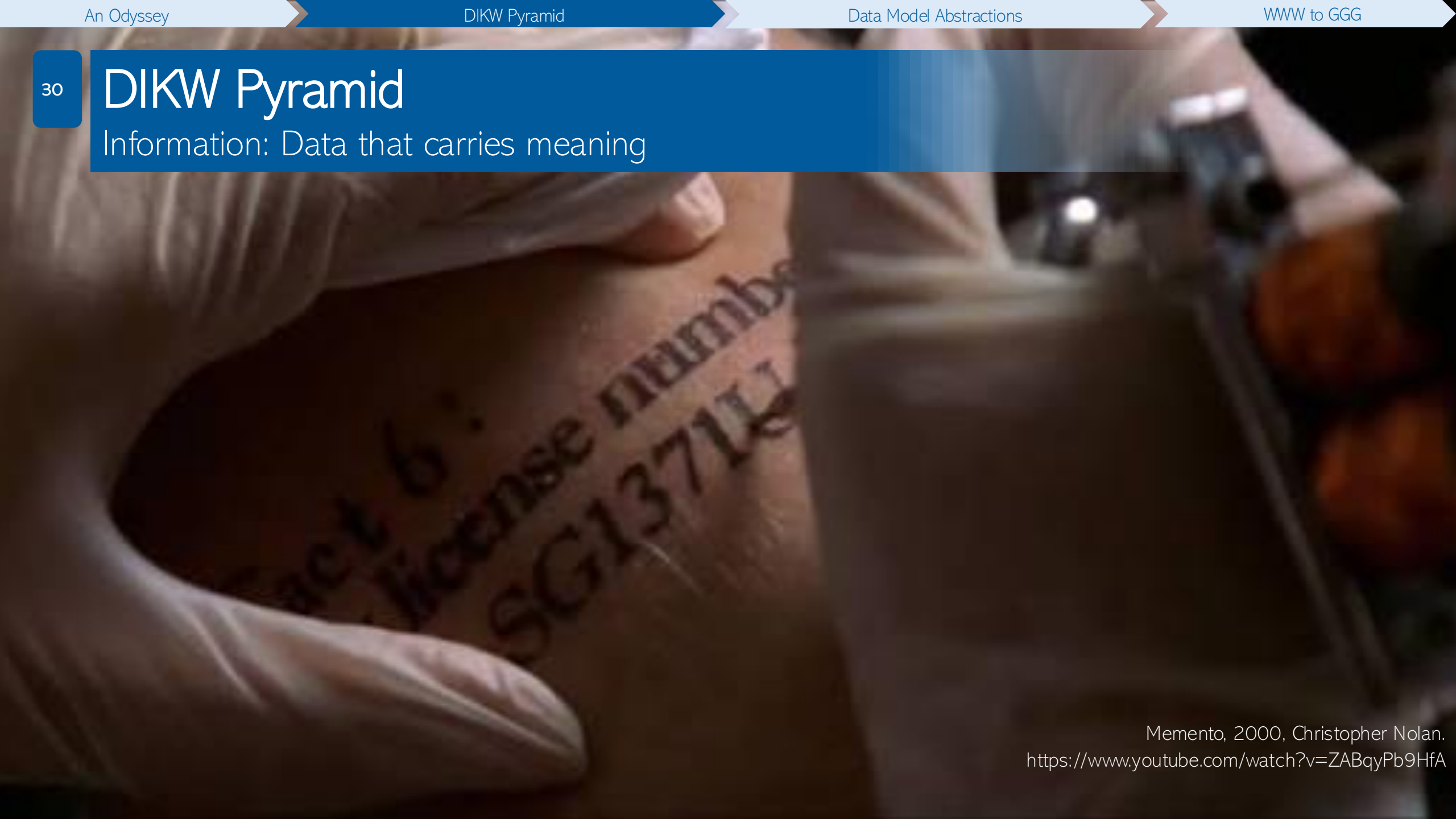
Example in Natural Language Processing



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DIKW Pyramid

Information: Data that carries meaning



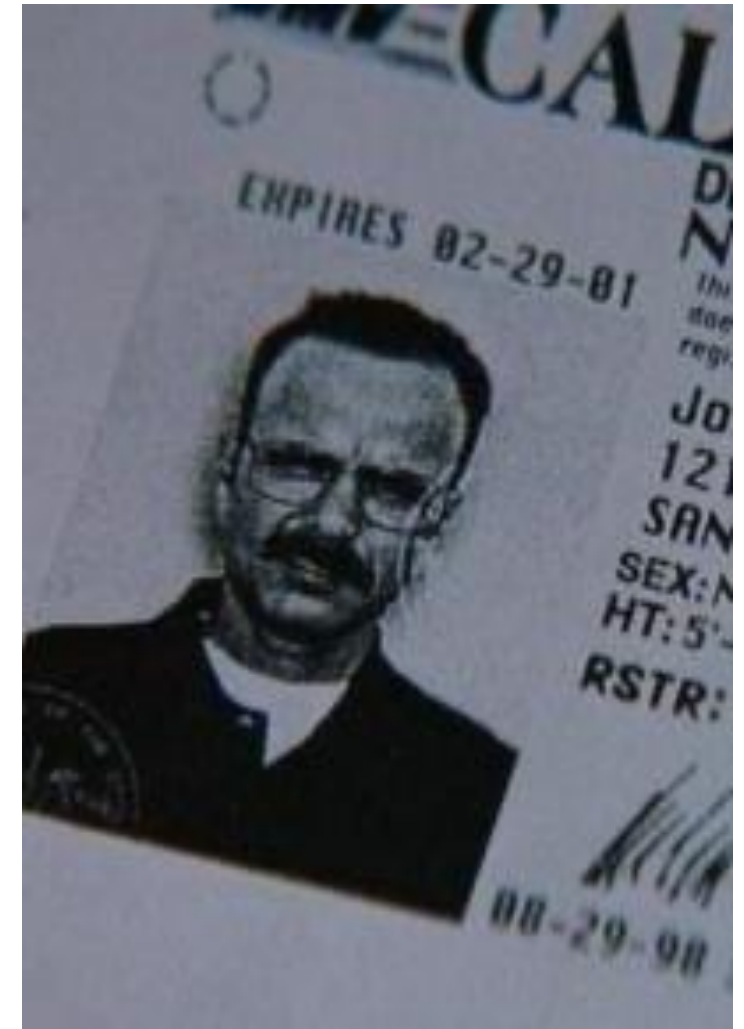
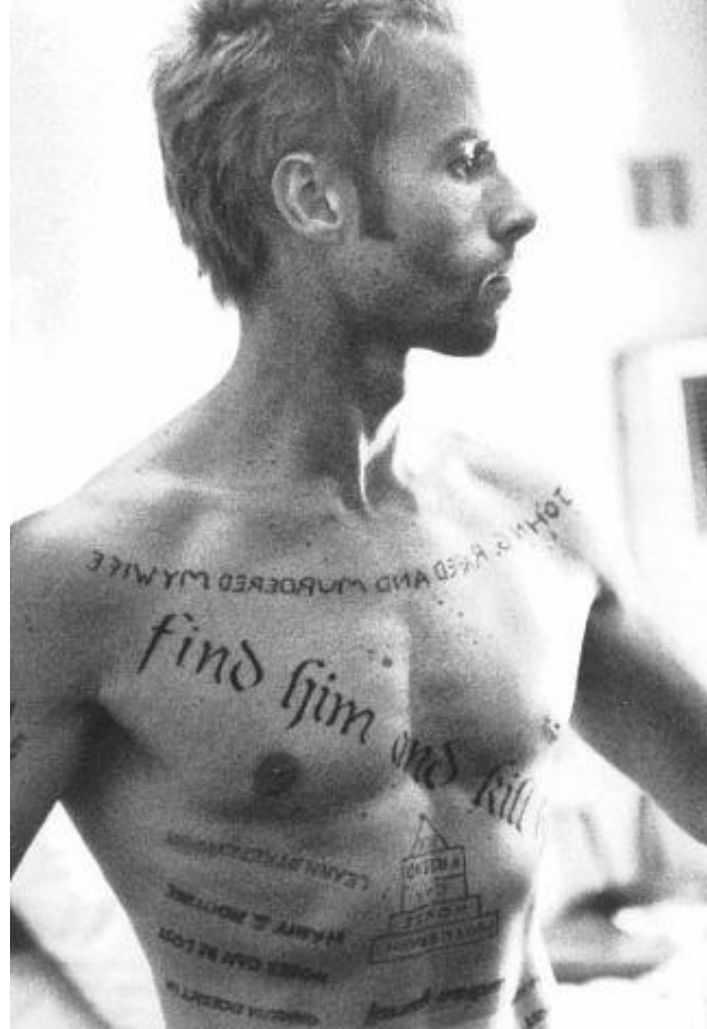
Memento, 2000, Christopher Nolan.
<https://www.youtube.com/watch?v=ZABqyPb9HfA>

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DIKW Pyramid

Knowledge: Judgment on Information

Wisdom: Act upon Knowledge



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Wisdom→Knowledge→Information→Data

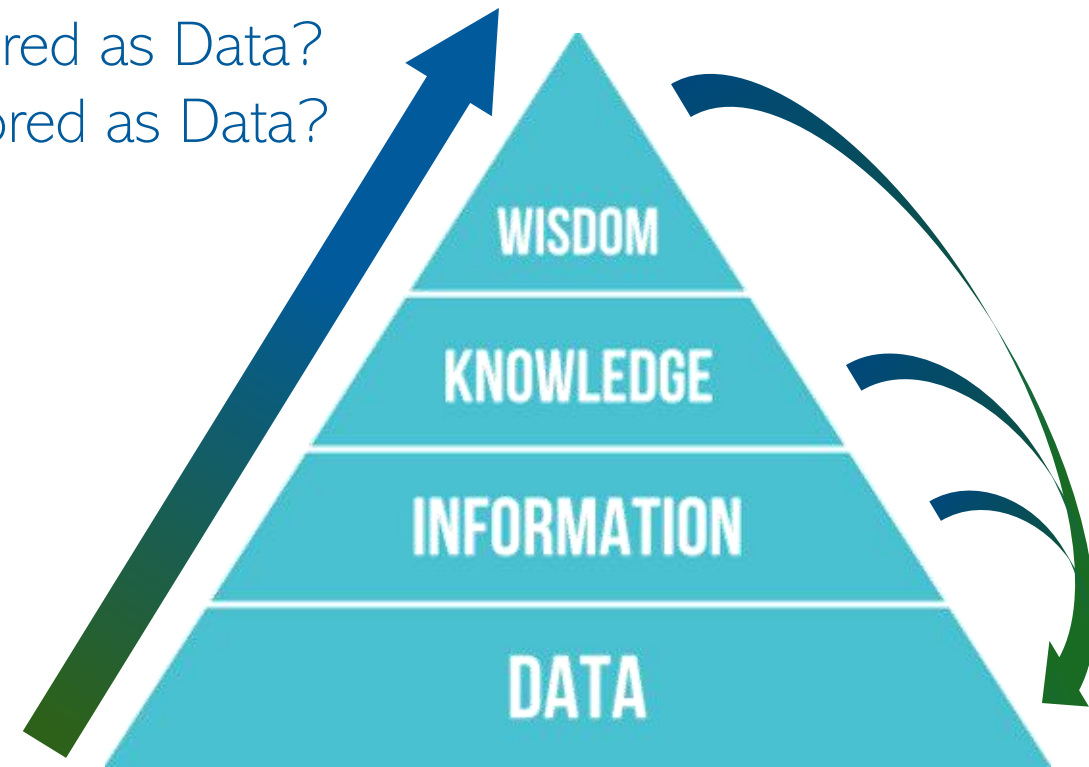
After the first button up, the result can be stored, but should it be stored like original raw data?

No!

What is an efficient model for **Wisdom** to be stored as Data?

What is an efficient model for **Knowledge** to be stored as Data?

What is an efficient model for **Information** to be stored as Data?



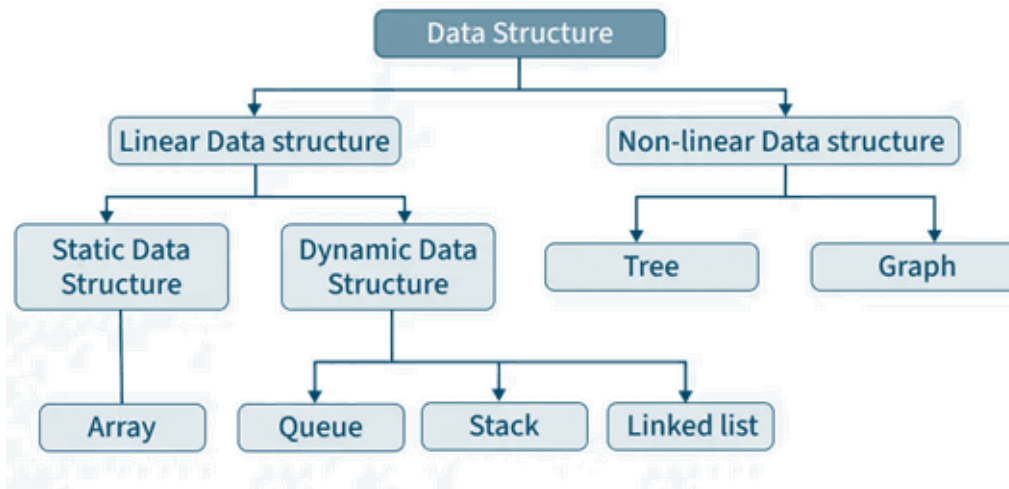
What is the difference? Efficient Reconstruction!

May or May not Relate to Real World

Social Network of People as Graph

Citation Network of Papers as Graph

B-Tree, R-Tree, ..., for Efficient Search



10 Data Structures Used in Daily Life

Data Structure	Illustration	Use Cases
List		Twitter feeds
Array		Math operations Large data sets
Stack		Undo/Redo of word editor
Queue		Printer jobs User actions in game
Heap		Task scheduling
Tree		HTML document AI decision
Suffix Tree		Search string in document
Graph		Friendship tracking Path finding
R-tree		Nearest neighbour
Hash Table		Caching systems

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DIKW Size

<https://www.martinhilbert.net/worldinfocapacity-html/>

Cannot be directly count ...

But by proxies like

recorded information,

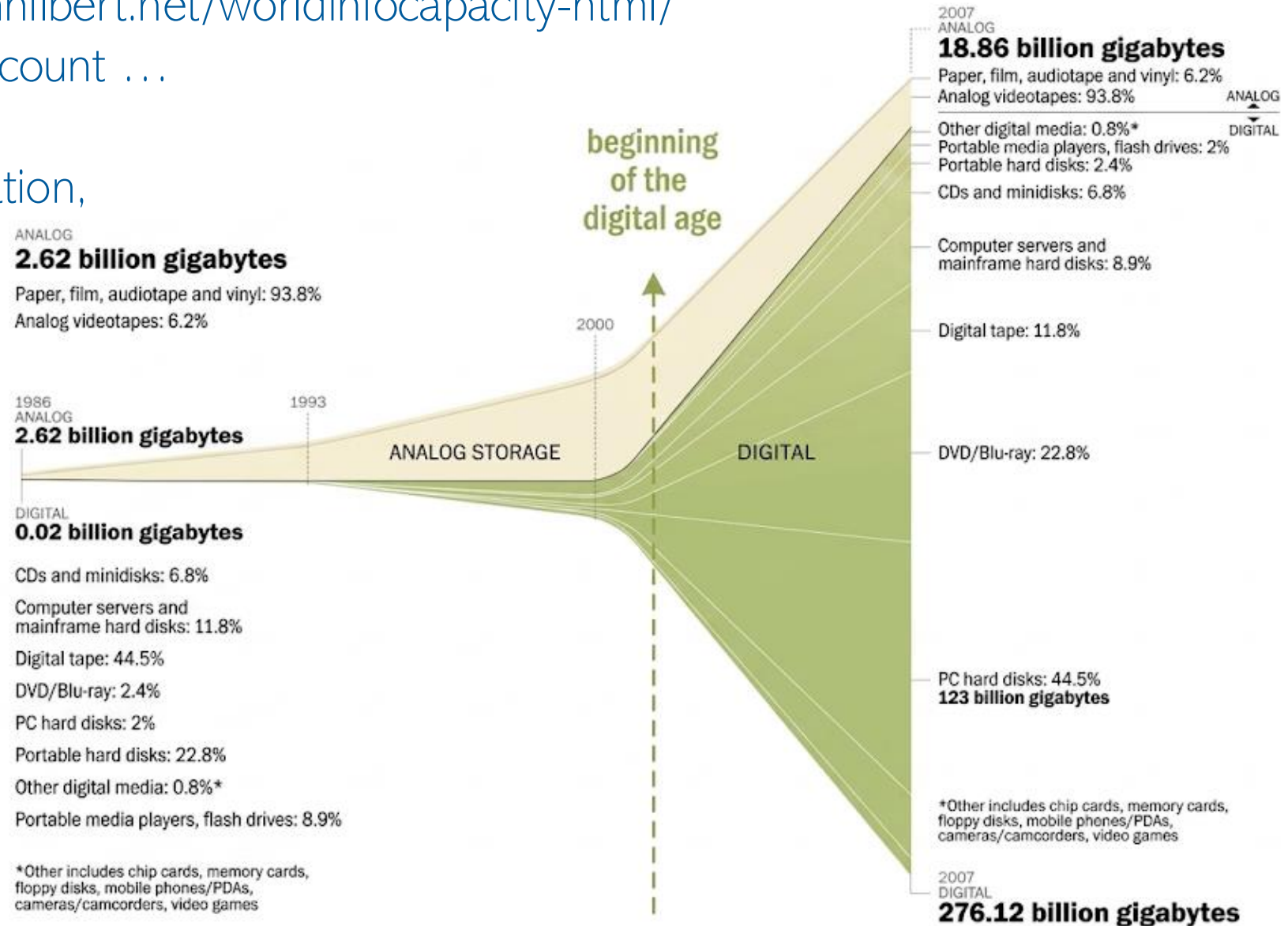
books,

publications,

documents,

webpages,

....



Outline: W01 B

I) An Odyssey

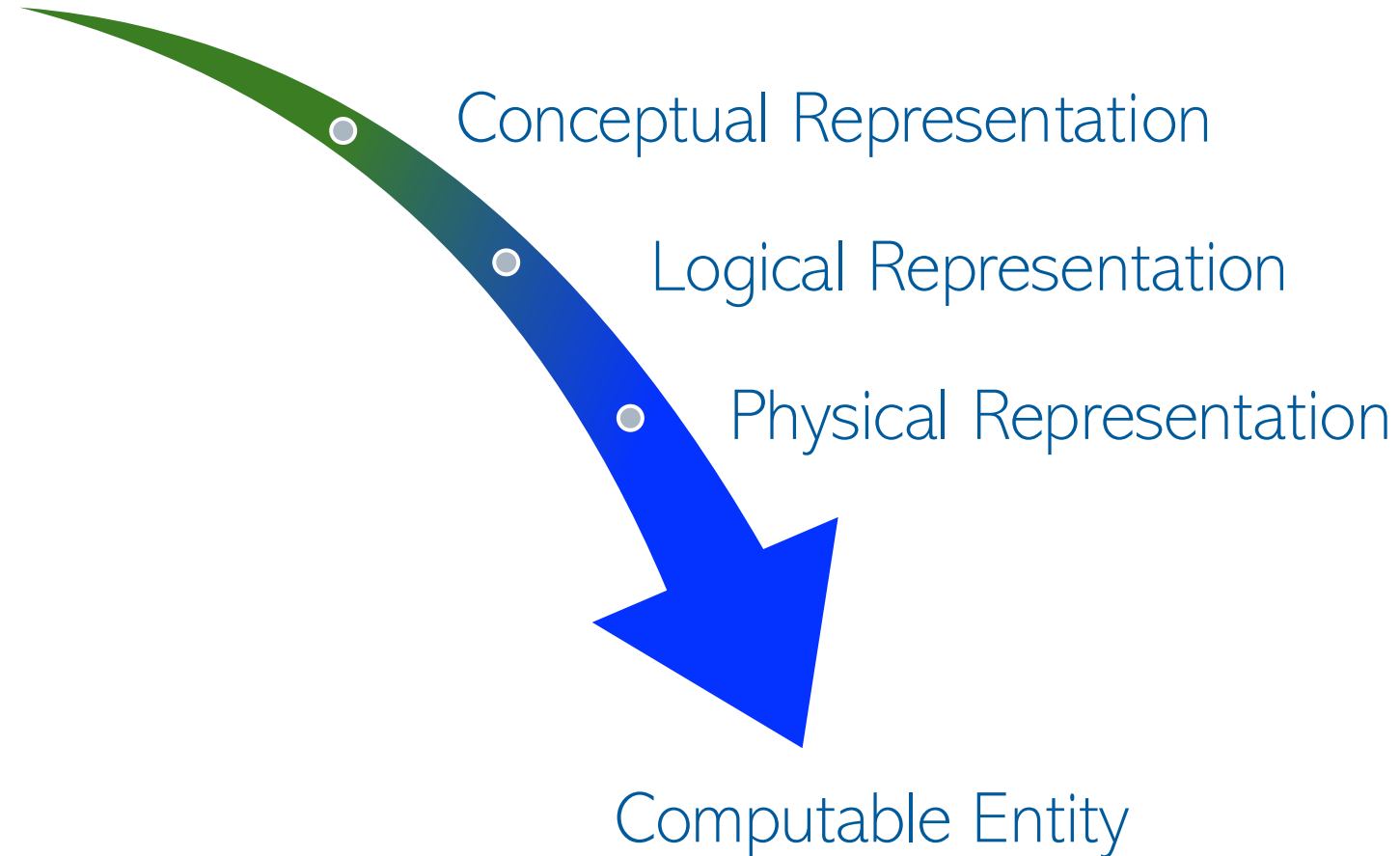
II) DIKW Pyramid

III) Data Model Abstractions

III) WWW to GGG

37 Data Model Abstractions

Real-World Observation



38 Data Model Abstractions

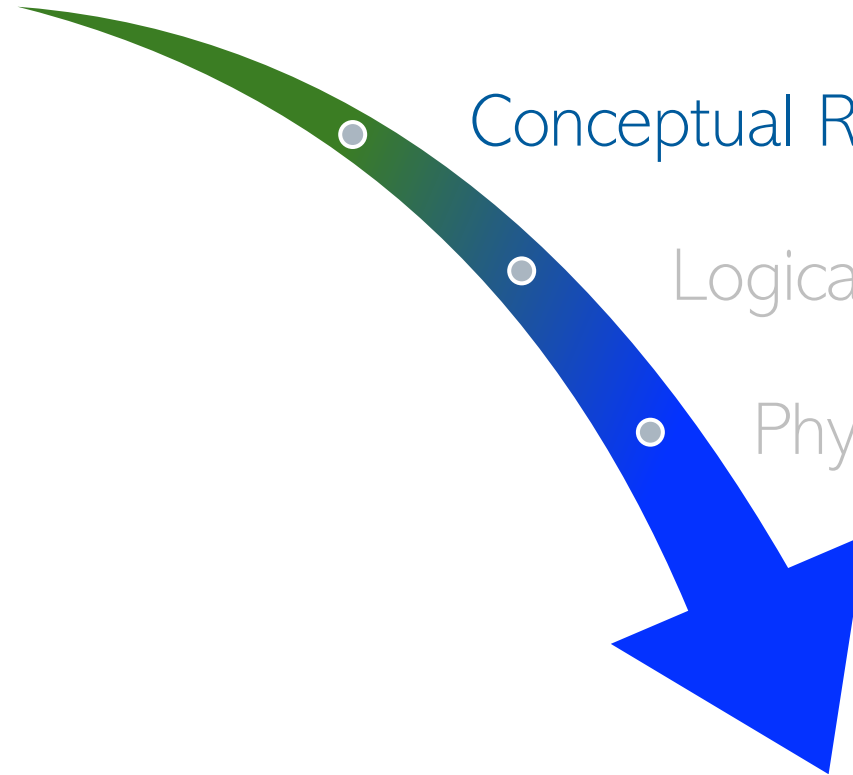
Real-World Observation

Conceptual Representation

Logical Representation

Physical Representation

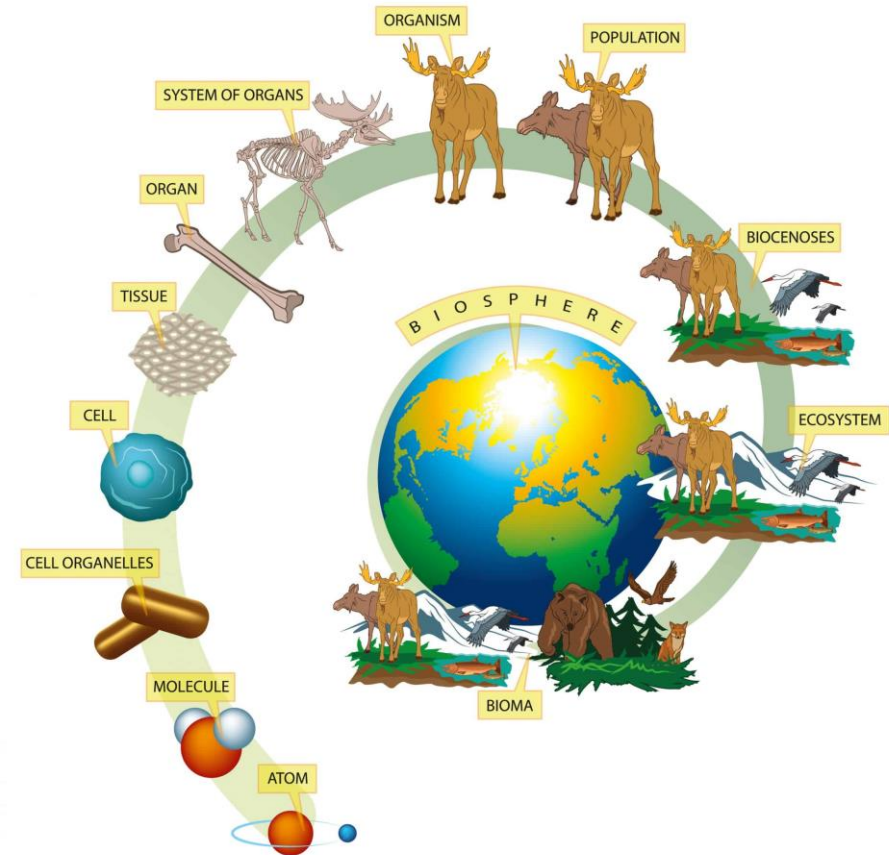
Computable Entity



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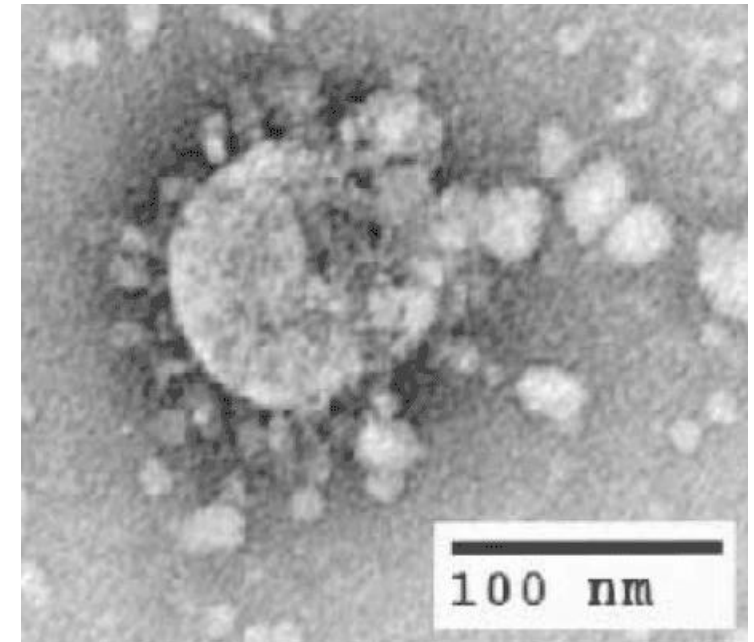
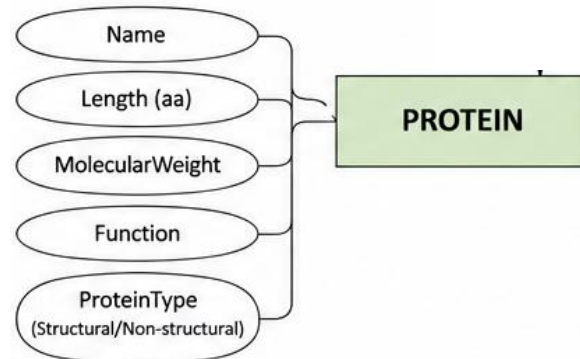
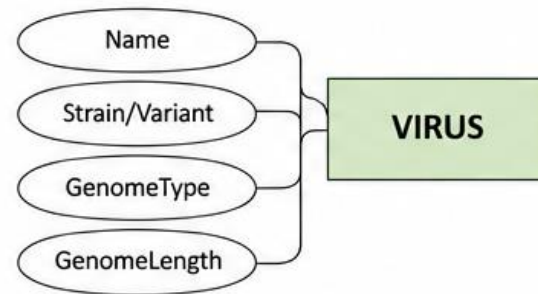
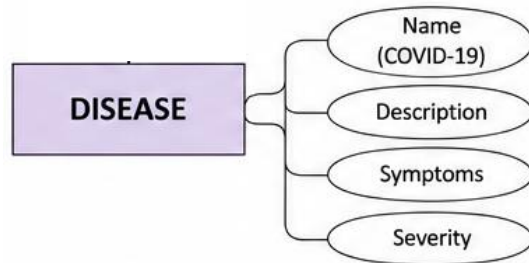
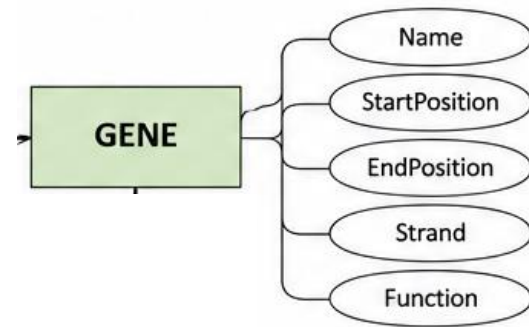
Abstractions: Conceptual Representation

1. Identify Real-World Entities, Attributes, Relationships
2. Create Schema



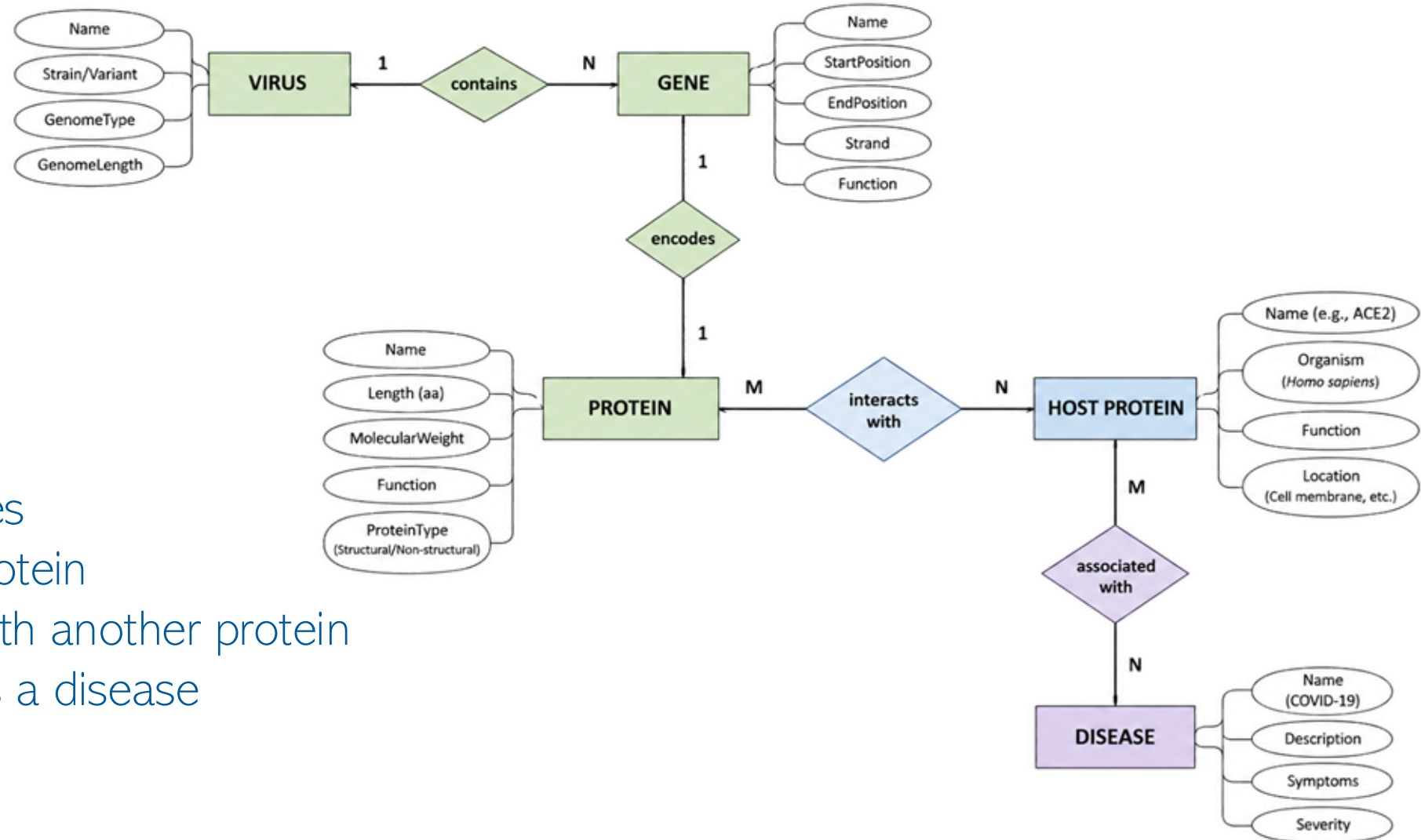
Abstractions: Conceptual Representation

Entities, Attributes



<https://en.wikipedia.org/wiki/SARS-CoV-2>

Abstractions: Conceptual Representation



Relationships

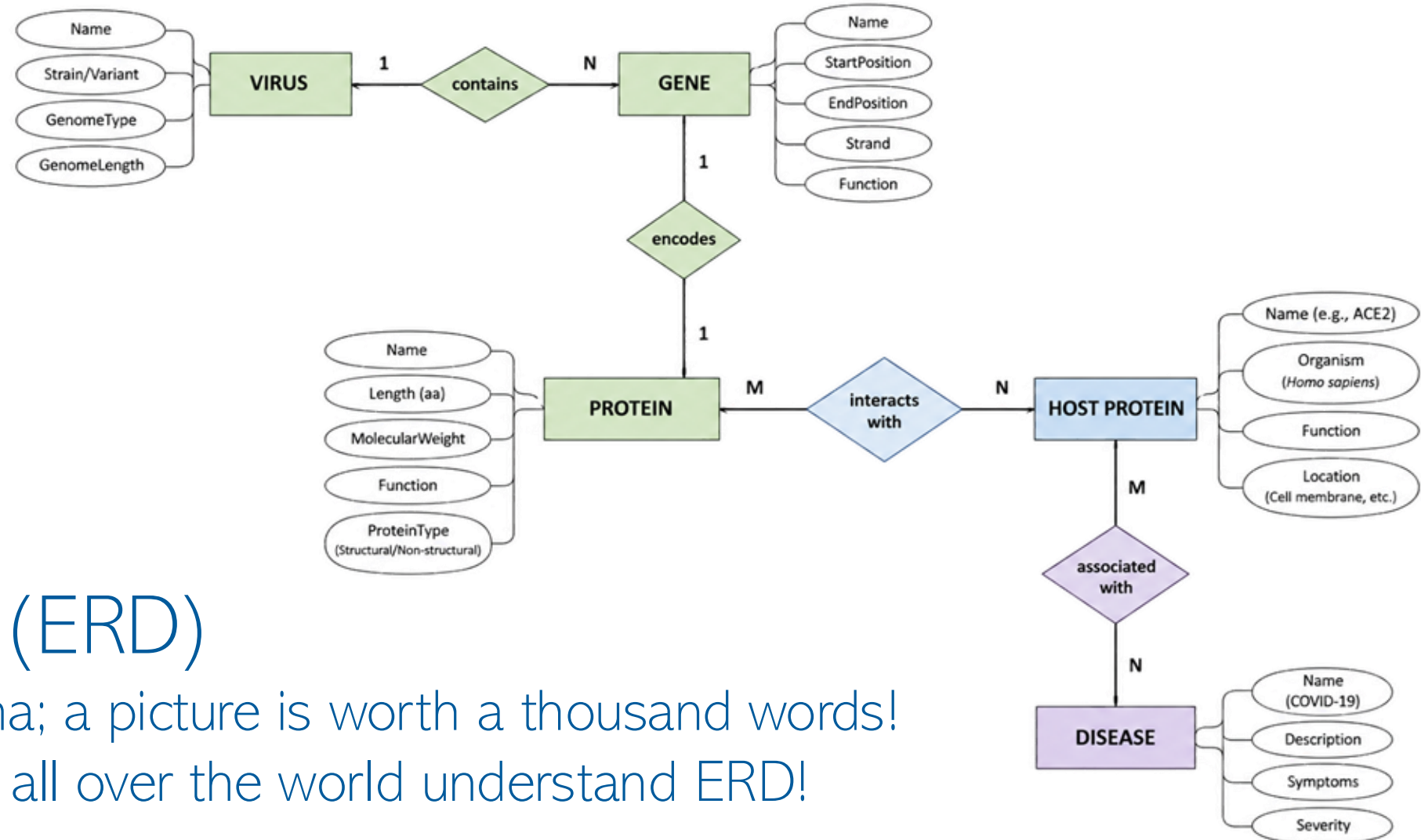
A virus **contains** genes

A gene **encodes** a protein

A protein **interacts** with another protein

An interaction **causes** a disease

Abstractions: Conceptual Representation



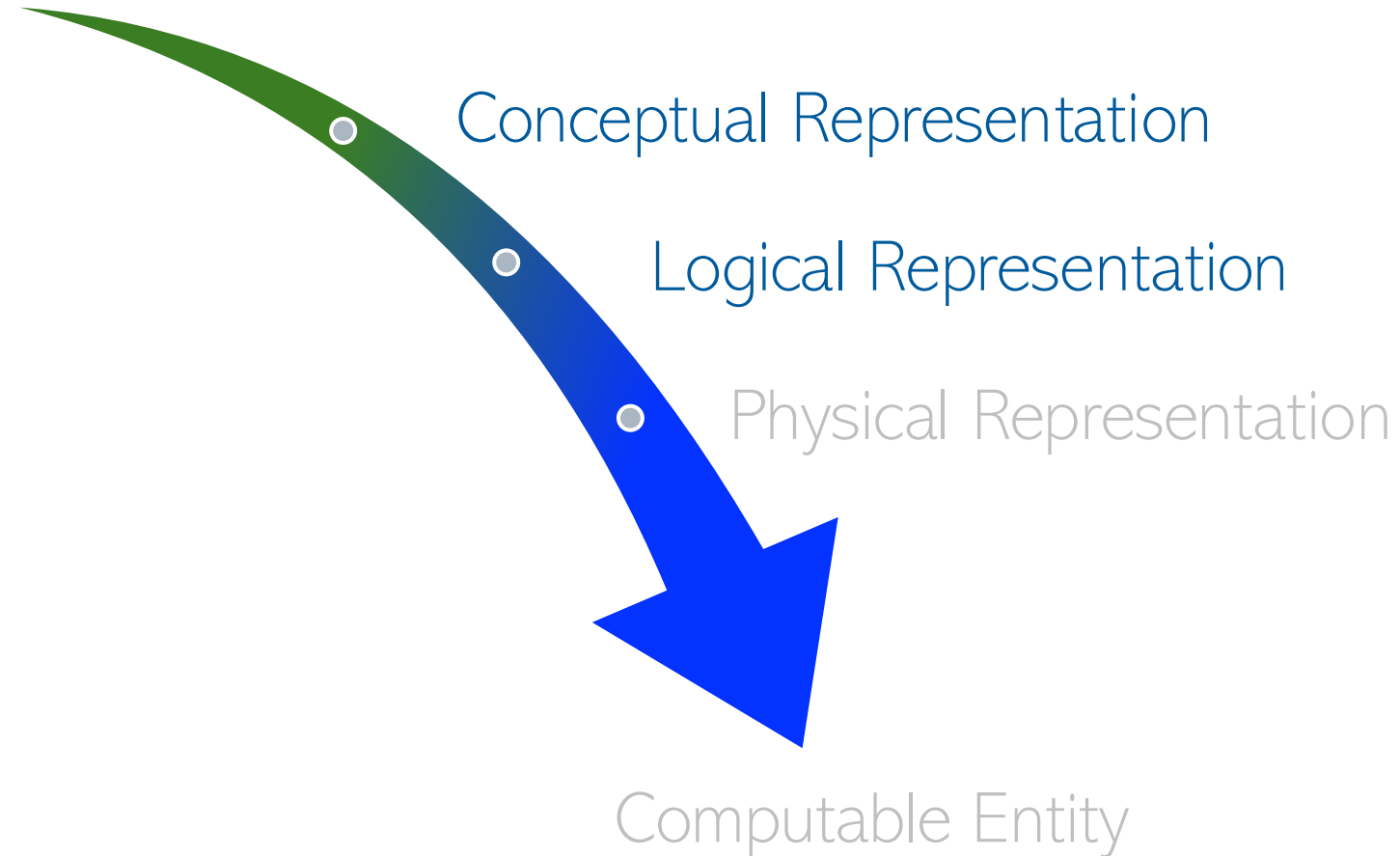
ER Diagram (ERD)

A graphical schema; a picture is worth a thousand words!

All data engineers all over the world understand ERD!

43 Data Model Abstractions

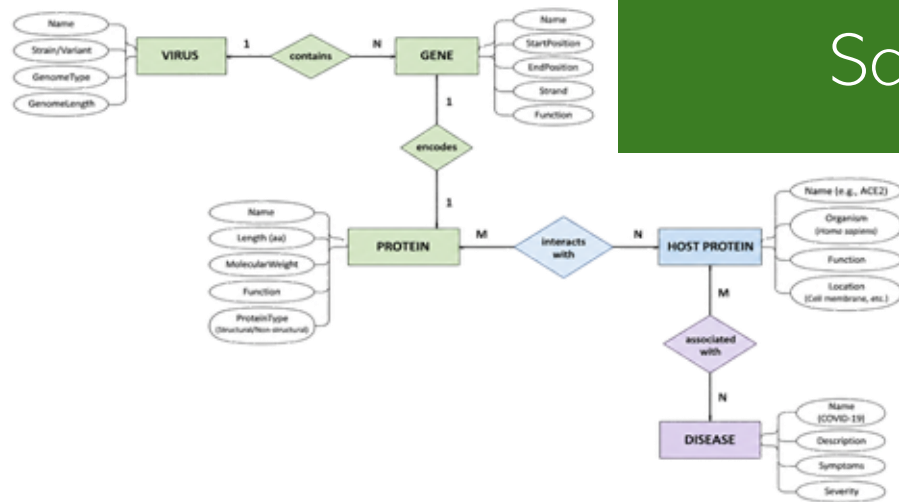
Real-World Observation



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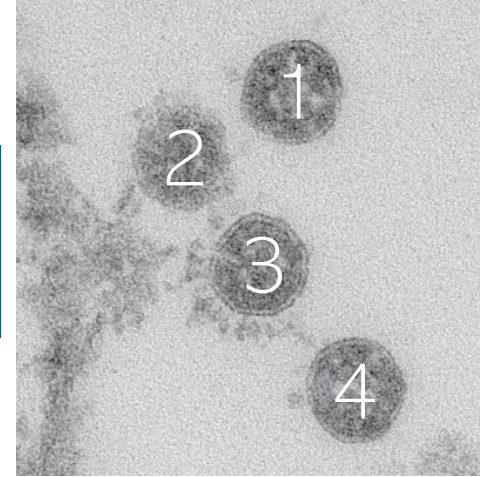
Abstractions: Logical Representation

How **instances** of entities, **values** of their attributes, their relationships should be represented based on a schema.



Schema

Real-World
Instances of Entity



Abstractions: Logical Representation

Data Schema

Data about Data: Meta-data

Defined at Setup Time

Rarely Change

Data Instance

Actual Data

Inserted at Running Time

Rapidly Change

Must Conform to Schema

Abstractions: Logical Representation

Tabular ...

Virus Entity			
Name (string)	Variant (string)	Genome Type (string)	Genome Length (decimal)
SARS-CoV-2	Wuhan-Hu-1	ssRNA	29,903 nt
Influenza	A/Puerto Rico/8/1934 (H1N1)	ssRNA	13,588 nt

Gene Entity				
Name (string)	Function (string)	Start (string)	End (decimal)	Strand (bool)
ORF1ab	Polyprotein / replication	266	21,555	+
ORF3a	Accessory protein	25,393	26,220	+

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Abstractions: Logical Representation

Tabular ...

Virus Entity			
Name	Variant	Genome Type	Genome Length

No Relationship!

Virus × Gene × Protein × Disease

Conceptual level identifies the entities, attributes, and relationships only.
Does not say how to represent them among actual instances!

ORF1ab	Polyprotein / replication	266	21,555	+
ORF3a	Accessory protein	25,393	26,220	+

Abstractions: Logical Representation

Tabular ...

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Abstractions: Logical Representation

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Redundancy! Wasting space. Not that important ...

Abstractions: Logical Representation

Tabular ...

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Name (string)	Variant (string)	Genome Type (string)	Genome Length (decimal)	Name (string)	Function (string)	Start (string)	End (decimal)	Strand (bool)
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				ORF3a	Accessory protein	25,393	26,220	+

Inconsistency! Incoherent pieces of information.

Abstractions: Logical Representation

Tabular ...

Virus Entity				Gene Entity				
Name (string)	Variant (string)	Genome Type (string)	Genome Length (decimal)	Name (string)	Function (string)	Start (string)	End (decimal)	Strand (bool)
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Speed! Cascading of updates/deletions.

Abstractions: Logical Representation

Not All (Data) Representations are Efficient.

Not All Conceptual Representations are Efficient.

Not All Logical Representations are Efficient.

Not All Physical Representations are Efficient.

Variety of representations have been proposed ...

1960: Object Oriented (Course: OOP using Java/C++/...)

1969: Relational: Mathematical Relationships (Course: DBMS)

1996: XML (This Course)

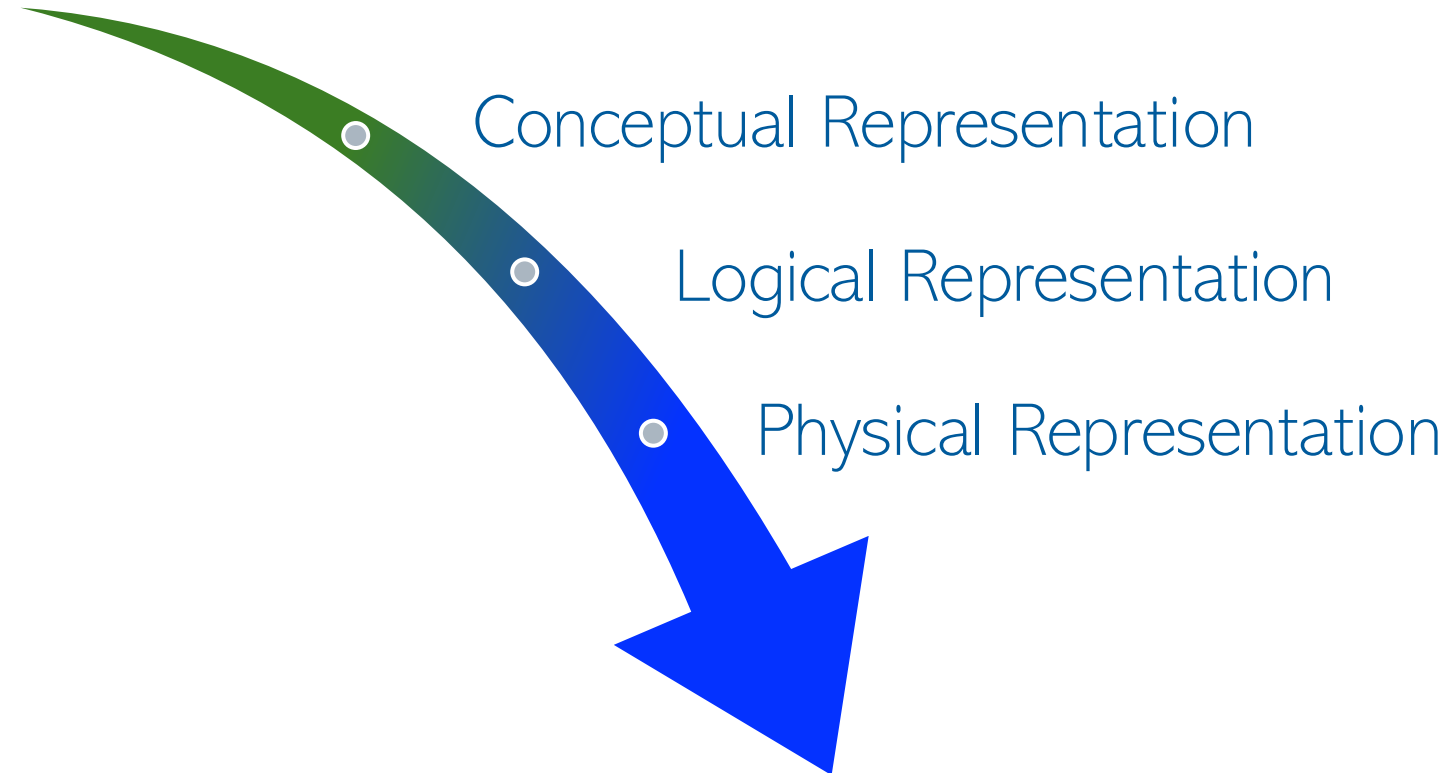
...

2007: Knowledge Graphs (This Course)

...

53 Data Model Abstractions

Real-World Observation



Computable Entity

Abstractions: Physical Representation

Where and how are the instances stored in physical storages.

Internet → Cloud → Database → File

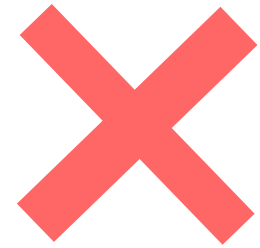


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Abstractions: Physical Representation

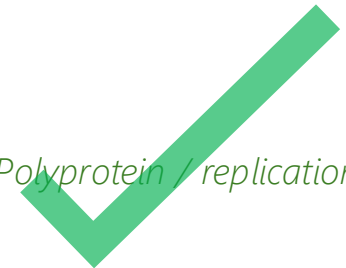
File: A 1D array (stack) of n-bit storage element

SARS-CoV-2, Wuhan-Hu-1, ssRNA, 29,903 nt
Influenza, A/Puerto Rico/8/1934 (H1N1), ssRNA, 13,588 nt
ORF1ab, Polyprotein / replication, 266, 21,555, +
ORF3a, Accessory, 25,393, 26,220, +



2D vs. 1D

SARS-CoV-2, Wuhan-Hu-1, ssRNA, 29,903 nt \n Influenza, A/Puerto Rico/8/1934 (H1N1), ssRNA, 13,588 nt \n ORF1ab, Polyprotein / replication, 266, 21,555,



JUST THE WAY
YOU ARE



Play (k)

0:24 / 3:56



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Abstractions: DataBase (Management) Systems (DBMS)

A Software System that provides

General Purpose,

Efficient,

Convenient,

Safe Multi-user Storage of and Access to
Massive Amounts of Data



Abstractions: DataBase Management Systems (DBMS)

Company	DBMS
Oracle	Oracle, MySQL
Microsoft	MySQLServer
IBM	DB2
Apache Software Foundation	CouchDB (NOSQL)

Outline: W01 B

I) An Odyssey

II) DIKW Pyramid

III) Data Model Abstractions

III) WWW to GGG

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Sir Timothy John Berners-Lee (1955-Present), aka TimBL

English Computer Scientist, Professorial Research Fellow at the University of Oxford, Professor Emeritus at the MIT

1989: World Wide Web (WWW), HTML, URL, HTTP

For human-readable publicly shared

Information

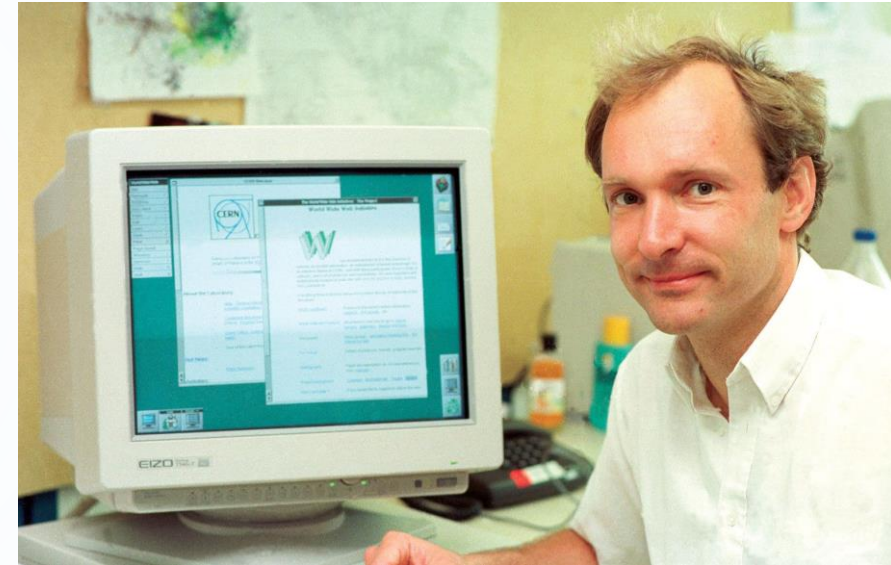
Knowledge

But proposed Giant Global Graph (GGG)

For machine-readable publicly shared

Information

Knowledge



Hyperlinks: W01 B

Constellation Fornax: <https://hubblesite.org/contents/media/videos/2014/27/766-Video.html>

Maya Civilization: https://en.wikipedia.org/wiki/Maya_numerals

Memento: <https://www.youtube.com/watch?v=ZABqyPb9HfA>

DIKW Size: <https://www.martinhilbert.net/worldinfocapacity-html/>
<https://pdodds.w3.uvm.edu/files/papers/others/2011/hilbert2011a.pdf>

Bruno Mars: <https://www.youtube.com/watch?v=LjhCEhWiKXk>